

Test-Time Warm-Up for Open-World Classification with Large Multimodal Models

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1 Introduction

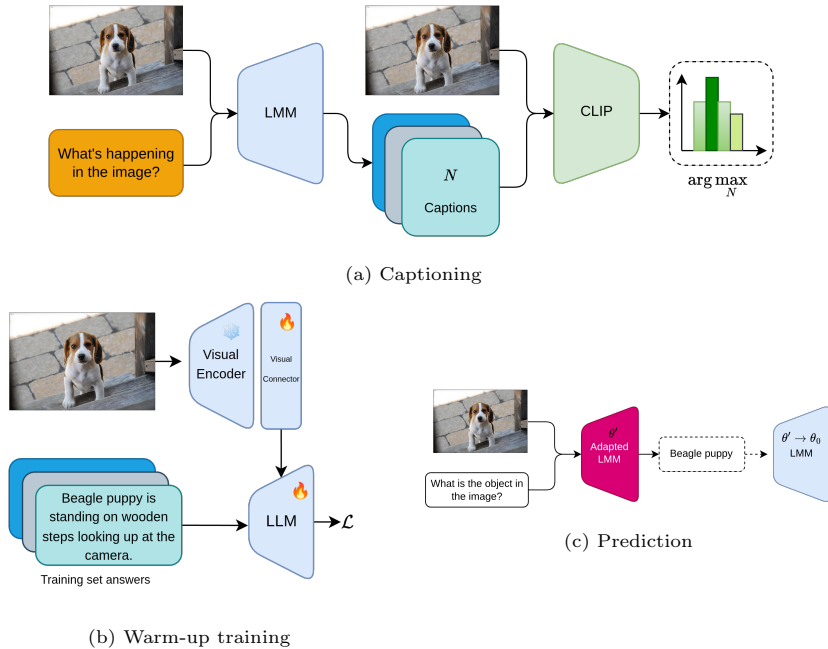


Figure 1: Detailed overview of Test-Time Warm-Up (TTW). For a given test image, the model first generates candidate captions from auxiliary prompts, and CLIP selects the best caption for each prompt. The model is then warm-up trained on the selected (prompt, caption) pairs for E epochs, updating only the chosen trainable parameters while the vision encoder remains frozen. In the prediction stage, the warmed-up model performs classification inference on the same image. After prediction, the adapted weights are restored to their pre-warm-up state, and the loop repeats independently for the next test image.

Image classification has traditionally been studied under the closed-world assumption, where the model must assign each image to one of a fixed, prede-

fixed set of labels. This assumption is convenient, but it does not match many real situations, where the set of possible categories is not known in advance. Large multimodal models (LMMs), which are trained autoregressively on images and text, offer a way to relax this assumption: instead of choosing from a fixed list, they can be prompted directly and generate a label as free-form text. This makes them naturally suited to the open-world setting, where the space of possible answers is essentially unbounded.

Recent work has begun to study how well LMMs actually perform in this setting. Conti et al. [4] introduce a benchmark to evaluate LMMs as open-world image classifiers, and follow-up work shows that providing a few in-context examples can substantially improve their accuracy, to the point of matching contrastive vision-language models such as CLIP [13] equipped with cache-based adapters [7]. These results suggest that LMMs already contain useful classification knowledge in their weights, and that the main challenge is finding good ways to bring this knowledge out at inference time, ideally without collecting labeled data for the target task.

In this work we explore a different way of doing this, based on test-time warm-up (TTW) [14]. This approach adapts the model to a single test image before making a prediction, and then discards the update to re-adapt the model to the next input sample. The adaptation does not require any labels: the model is prompted with a set of generic auxiliary questions about the image (for example, “what objects are visible in this image?”), its own answers are filtered using CLIP to keep the most reliable ones, and a few gradient steps are taken on these filtered answers. The intuition is that answering simple questions about the image first “warms up” the model and makes it pay closer attention to the visual details, so that it is better prepared for the actual classification question. TTW was originally proposed for visual question answering, and to our knowledge it has not been tested on open-world image classification.

The goal of this project is to reproduce test-time warm-up and to study how it behaves on the open-world classification benchmark of Conti et al. [4]. Concretely, our contributions are the following:

- We reproduce the test-time warm-up methodology and apply it to LMMs in the open-world image classification setting.
- We evaluate the method on the benchmark of Conti et al. [4] and compare it against the baseline reported in their work.
- We explore a GRPO post-training strategy [20] as a preliminary warm-start before test-time warm-up, using reinforcement learning to encourage structured visual reasoning before classification.

The implementation for this project is available at: <https://github.com/jucamohedano/lmms-ocw/tree/ttw>.

2 Related works

LMMs as image classifiers. Image classification has traditionally relied on the closed-world assumption, in which every test image is assigned to one of a fixed set of categories. Contrastive vision-language models such as CLIP [13] dominate this setting through zero-shot prompting. Large multimodal models (LMMs), which instead generate free-form text, are a natural fit for the open-world setting, where the label is not restricted to a predefined list. Conti et al. [4] formalize this open-world classification setting for LMMs and introduce an evaluation protocol that goes beyond exact label matching, using complementary text- and semantic-alignment metrics. Their benchmark compares 13 autoregressive LMMs across 10 datasets with contrastive vision-language alternatives, showing that LMMs are stronger than contrastive retrieval-style methods in the open-world setting, but still trail closed-world CLIP classifiers that are allowed to use a fixed label set. Their analysis further identifies recurring error modes, especially overly generic predictions and failures on fine-grained distinctions, and shows that prompting and reasoning can reduce some of these errors. Building on these conclusions, Garosi et al. [7] study LMMs as in-context classifiers: in the closed-world setting, they show that LMMs given enough labelled in-context examples (up to 16 shots) can match or surpass contrastive models equipped with cache-based adapters; in the open-world setting, they observe that naive in-context learning does not reliably help, and propose CIRCLE, an iterative strategy that refines the context, to obtain improvements over the open-world baselines. Our work follows this line of building open-world classifiers on top of LMMs, but instead of conditioning a frozen model on in-context examples, we ask whether adapting the model’s parameters at test time can improve open-world classification on the benchmark of Conti et al.

Test-time adaptation. Test-time adaptation (TTA) updates a model during inference using only the unlabeled test input, and resets it afterwards. The idea originates in Test-Time Training (TTT) [17], which turns each test image into a self-supervised task (rotation prediction) and takes a few gradient steps before predicting. TENT [18] instead adapts by minimizing the entropy of the model’s predictions. For vision-language models, this entropy-minimization idea was adapted by TPT [15], which tunes CLIP’s text prompt per test image, and by MEMO [19], which adapts from a single image using augmented views. A separate, training-free line avoids gradient updates altogether: TDA [10] keeps a lightweight cache of confident predictions and refines new predictions against it. Notably, ZERO [5] questions whether the gradient step in entropy-minimization methods is doing useful work at all: it shows theoretically and empirically that minimizing the marginal entropy largely leaves the predicted class unchanged, and that simply augmenting the image, keeping the confident views, and voting after setting the softmax temperature to zero already matches or beats prompt tuning, at a fraction of the cost. Test-time warm-up belongs to this broad family but differs in two important ways. First, it adapts a generative LMM rather than a contrastive VLM. Second, rather than minimizing prediction entropy

or voting over augmentations, it takes explicit gradient steps on self-generated, CLIP-filtered auxiliary captions by minimizing the autoregressive negative log-likelihood (cross-entropy) of the caption tokens, which keeps it applicable to the open-world setting where the closed-set assumptions behind methods like TDA and ZERO no longer hold.

Test-time training for language and reasoning models. The same idea has also been applied in the language domain. Hardt and Sun [8] fine-tune an LLM on the retrieved nearest neighbours of each test input before predicting, and Hu et al. [9] adapt an LLM to a target domain from unlabeled test data using a perplexity-based objective together. Most relevant to us, Akyürek et al. [1] show that per-instance test-time training with auxiliary tasks and augmentations yields large gains on abstract reasoning, and identify per-instance updates and auxiliary-task design as the key ingredients. Test-time warm-up [14] can be seen as transferring this recipe to the multimodal setting, using auxiliary visual prompts to “warm up” the LMM before it answers.

Reinforcement learning for open-world classification. Recent work has also explored reinforcement learning for open-world image classification. SpeciaRL [2] finetunes reasoning LMMs with GRPO using a dynamic verifier-based reward, with the goal of making fine-grained predictions more specific while preserving correctness. Our GRPO phase is related, but we use it as a preliminary warm-start before test-time warm-up.

3 Methodology

Test-time warm-up adapts the model separately for every test image, using a short warm-up step right before the model is asked to classify the image. The key idea is that the model is not trained on the classification labels themselves. Instead, it is trained for a few steps on simple descriptive captions of the image, which it generates on its own. After the model has classified the image, the warm-up update is undone, so the next image always starts from the same original model. Following the original method of Rajaneesh et al. [14], we apply this idea to open-world classification on the LMMS-OCW benchmark of Conti et al. [4].¹

3.1 Overview of the warm-up loop

For each test image x , we first build a small warm-up set

$$\mathcal{D}_x = \{(x, q_i, c_i)\}_{i=1}^N, \quad (1)$$

where q_i is an auxiliary prompt and c_i is the CLIP-selected caption for that prompt, as described in Section 3.4. Before adapting the model, we snapshot

¹<https://github.com/altndrr/lmms-owc>

the trainable parameters θ_0 . Warm-up then performs T gradient steps on the caption-token negative log-likelihood, where T is the total number of optimization steps induced by the chosen number of epochs and mini-batches,

$$\theta_x = \text{TTW}(\theta_0, \mathcal{D}_x) = \theta_0 - \eta \sum_{t=1}^T \nabla_{\theta} \mathcal{L}_{\text{cap}}(\theta_{t-1}; \mathcal{D}_x), \quad (2)$$

with

$$\mathcal{L}_{\text{cap}}(\theta; \mathcal{D}_x) = - \sum_{(x,q,c) \in \mathcal{D}_x} \sum_{m=1}^{|c|} \log p_{\theta}(c_m \mid x, q, c_{<m}). \quad (3)$$

The prediction for the image is then generated with the adapted parameters,

$$\hat{y} = f_{\theta_x}(x, q_{\text{cls}}), \quad (4)$$

where q_{cls} is the classification prompt. After this prediction, we discard θ_x and restore θ_0 before processing the next image. Thus each image is adapted independently, the model never accumulates updates across test images, and any degradation of general knowledge caused by one warm-up step is undone before the next input.

3.2 Base model

All experiments use Qwen vision-language models, specifically Qwen2-VL-7B-Instruct and Qwen2.5-VL-7B-Instruct. These models pair a visual encoder with a Qwen language-model backbone rather than using a single monolithic decoder-only transformer. At a high level, they contain three main parts: a vision encoder that turns the image into visual features, a vision-language connector or merger that maps these features into the language model’s embedding space, and the language model that produces the text answer. During warm-up we always keep the vision encoder frozen, and we always allow the connector or merger to be trained. The exact subset of language-model parameters or adapters updated during warm-up depends on the finetuning method described below.

3.3 Finetuning methods

We compare three ways of adapting the model during warm-up. They differ in which parameters are trainable, and therefore in their memory cost. In all cases, the vision encoder is frozen, the connector or merger is trainable, and the optimization loss is the caption-token causal language-modelling cross-entropy defined above: prompt and image tokens are masked out, so the model is trained to produce the captions rather than to repeat the prompt.

Dense finetuning. Dense finetuning updates the language-model weights directly, together with the connector or merger parameters. If θ_{LM} denotes the

language-model weights and θ_{conn} the connector or merger parameters, the trainable set is

$$\Theta_{\text{full}} = \{\theta_{\text{LM}}, \theta_{\text{conn}}\}. \quad (5)$$

This is the most flexible adaptation strategy, but also the most memory-intensive because gradients and optimizer state must be stored for the full language model.

LoRA finetuning. LoRA keeps the original language-model weights frozen and adds trainable low-rank updates to selected attention and feed-forward matrices. For a frozen weight matrix W_0 , LoRA uses

$$W = W_0 + \Delta W, \quad \Delta W = BA, \quad \text{rank}(\Delta W) \leq r, \quad (6)$$

where A and B are trainable low-rank matrices and we use rank $r = 16$. The trainable set is therefore

$$\Theta_{\text{LoRA}} = \{A, B, \theta_{\text{conn}}\}, \quad (7)$$

with separate adapter matrices for each target layer. This is much lighter than dense finetuning because the dense base weights remain fixed.

Singular value finetuning. Singular value finetuning (SVF) first decomposes each selected weight matrix as

$$W_0 = U\Sigma V^\top, \quad (8)$$

where U and V contain orthonormal directions and the diagonal matrix Σ contains the singular values. During warm-up, SVF keeps U and V fixed and trains only the singular values,

$$W = U\Sigma'V^\top, \quad \Theta_{\text{SVF}} = \{\Sigma', \theta_{\text{conn}}\}. \quad (9)$$

Thus SVF adapts the model by rescaling existing directions in each weight matrix rather than learning new directions. This makes it even lighter than LoRA. We use full-rank SVF ($\text{rank}(\Sigma')$ equals the original rank of each matrix), so no singular directions are discarded during decomposition.

3.4 Warm-up captions

The warm-up captions are short, generic descriptions of the image, and the model generates them itself, so no human labels are needed. We use the same ten auxiliary prompts as the original method, listed in Appendix B.

For each prompt the model generates several candidate captions, and a CLIP model scores how well each candidate matches the image; we keep the best-scoring caption per prompt, giving ten (prompt, caption) pairs per image. As the method’s name (*warm-up*) suggests, these captions can be produced online, right before classifying each image, as illustrated in Figure 1. In practice we generate them *offline* and store them on disk to save time during evaluation, and fall back to online generation whenever a caption is missing. The generation hyperparameters are listed in Appendix E.

3.5 Evaluation

We evaluate on the LMMS-OCW benchmark [4], which tests both vision-language models and large multimodal models on fine-grained open-world classification. We do not use the full benchmark. Instead, we evaluate on five of its ten datasets, spanning three of its four categories: Caltech101 [6], DTD [3], Flowers102 [11], Oxford Pets [12], and UCF101 [16]. These five datasets cover general objects, textures, human actions, flowers, and pets. Their sizes, class counts, and category groupings are listed in Appendix A, Table 5.

At test time each image is given a single classification question, and the model answers with free-form text. Because the answer is free text rather than a choice from a fixed list, we do not score it with plain accuracy. Instead, we use the benchmark metrics introduced by Conti et al. [4]: Text Inclusion (TI) checks whether the ground-truth label appears in the prediction; Semantic Similarity (SS) measures how close the prediction and the label are in meaning; and Concept Similarity extracts the individual concepts from the free-form answer and compares them to the label, reported both as the best match (bCS) and the median match (mCS). Following Garosi et al. [7], we report mCS alongside bCS because bCS alone can be inflated by answers that list many candidate concepts. Higher is better for all metrics.

Each image is processed on its own, i.e., batch size=1, which matches the per-image nature of the warm-up. We fix the random seed to 30 across all experiments. We compare three warm-up configurations against the zero-shot baseline of Conti et al. [4]: dense-finetuning warm-up, LoRA warm-up, and SVF warm-up. As a preliminary extension, we also evaluate a variant that first applies a GRPO post-training phase.

3.6 GRPO post-training phase

As an additional and still exploratory phase, we test whether a reinforcement-learning warm-start before test-time warm-up can help. We use Group Relative Policy Optimization (GRPO) [20], a reinforcement-learning algorithm first introduced for mathematical reasoning in language models. GRPO is a variant of Proximal Policy Optimization that removes the need for a separate value network: for each input it samples a group of candidate outputs from the current policy, scores each one with a reward function, and converts the rewards into relative advantages using the mean and standard deviation within the group, so that higher-reward outputs are encouraged relative to lower-reward ones. A KL penalty keeps the updated policy close to a reference model. Because the baseline comes from the group itself rather than a learned critic, GRPO is comparatively cheap to run.

We use GRPO to optimize two objectives at the same time: answer correctness and output structure. The structure objective encourages the model to first describe visible properties, parts, and setting inside a structured `<think>` block before naming the class. Our hypothesis is that this combination can make the model both more accurate and more consistent, while also producing richer

auxiliary captions during the later warm-up phase. Applying GRPO to open-world classification follows recent work in fine-grained open-world classification. SpeciaRL [2] fine-tunes reasoning LMMs with GRPO using a dynamic verifier-based reward. The goal is to make predictions more specific while preserving correctness.

Reward function. Our reward is static and rule-based: an output o receives $R(o) = R_{\text{fmt}}(o) + R_{\text{ans}}(o) \in [0, 1]$, combining a format score (up to 0.3) and an answer score (up to 0.7). The answer score is binary, rewarding outputs whose text after `</think>` contains the ground-truth label y^* as a whole word:

$$R_{\text{ans}}(o) = \begin{cases} 0.7, & \text{if } y^* \text{ matches the answer,} \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

The format score rewards well-formed reasoning through a gated rule-based function. Let $\mathcal{T} = \{\text{HASPROPERTY}, \text{HASA}, \text{ATLOCATION}\}$ be the required tags, and let $G(o) \in \{0, 1\}$ indicate whether o matches a `<think>...</think>` block containing all tags in $\mathcal{T}$ with non-empty contents. If $G(o) = 0$, the format reward is zero. Otherwise, for each tag $t \in \mathcal{T}$, let $E_t(o)$ be the comma-separated entries extracted from that tag. We define

$$R_{\text{fmt}}(o) = 0.3 G(o) \frac{1}{3} \left(\frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} M_t(o) + \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} P_t(o) + \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} D_t(o) \right), \quad (11)$$

where for each tag $t \in \mathcal{T}$,

- $M_t(o) \in \{0, 1\}$ is the **min-content** gate: 1 if $E_t(o)$ contains at least two entries and every entry has at least two characters, and 0 otherwise;
- $D_t(o) \in \{0, 1\}$ is the **deduplication** gate: 1 if no entry appears more than once within tag t , and 0 otherwise;
- $P_t(o) \in [0, 1]$ is the **POS-sanity** score, defined as the fraction of entries in $E_t(o)$ that pass a per-entry part-of-speech check.

Each entry is parsed with spaCy (`en_core_web_sm`) and fails the check if it spans more than three tokens or if any of its tokens is a verb or auxiliary. Otherwise, the entry passes if its syntactic root is an adjective, noun, or proper noun for HASPROPERTY, and a noun or proper noun for HASA and ATLOCATION. The per-tag score $P_t(o)$ is simply the average of these binary pass fail outcomes across all entries in $E_t(o)$. No external knowledge base is needed.

4 Results

Table 1 reports the main open-world classification results for Qwen2-VL-7B-Instruct. We compare the zero-shot baseline with dense finetuning, LoRA, and SVF warm-up, using the evaluation protocol described above.

Table 1: Open-world results on five LMMS-OCW datasets with Qwen2-VL-7B, grouped by category. We report Text Inclusion (TI), Semantic Similarity (SS), best Concept Similarity (bCS), and median Concept Similarity (mCS). Higher is better on all metrics; **bold** marks the best method per dataset and metric.

Dataset	Method	TI	SS	bCS	mCS
<i>Prototypical</i>					
Caltech101	Zero-shot	63.2	55.7	81.2	51.4
	TTW full	63.8	55.6	81.7	50.7
	TTW LoRA	64.6	53.8	81.7	47.2
	TTW SVF	64.9	53.4	82.4	46.4
<i>Non-prototypical</i>					
DTD	Zero-shot	15.8	28.5	50.3	27.8
	TTW full	16.6	28.7	50.8	28.0
	TTW LoRA	17.0	29.1	51.9	27.8
	TTW SVF	17.1	29.3	52.4	28.0
UCF101	Zero-shot	12.3	43.1	58.9	38.9
	TTW full	12.8	43.6	60.0	38.8
	TTW LoRA	16.1	45.5	63.2	35.7
	TTW SVF	13.8	44.6	61.8	37.7
<i>Fine-grained</i>					
Flowers102	Zero-shot	40.2	40.9	67.6	32.9
	TTW full	44.3	42.9	70.3	33.4
	TTW LoRA	50.5	46.3	73.0	33.8
	TTW SVF	50.3	45.9	73.3	32.8
Oxford Pets	Zero-shot	12.3	25.1	43.2	25.2
	TTW full	16.3	26.4	45.5	26.4
	TTW LoRA	47.0	35.9	65.3	32.2
	TTW SVF	42.5	34.7	61.8	32.6

Across TI, SS, bCS, and mCS, test-time warm-up improves over the zero-shot baseline most consistently on the datasets where the baseline is weakest, while the two parameter-efficient variants (LoRA and SVF) give the largest gains. The clearest improvements are on the two fine-grained datasets. On Oxford Pets, LoRA gives the largest gains in TI (12.3 to 47.0), SS (25.1 to 35.9), and bCS (43.2 to 65.3), while SVF gives the best mCS (25.2 to 32.6). On Flowers102, LoRA gives the best TI, SS, and mCS improvements (40.2 to 50.5, 40.9 to 46.3, and 32.9 to 33.8, respectively), while SVF gives the best bCS improvement (67.6 to 73.3). On datasets where the baseline is already strong, such as Caltech101 (bCS above 81 and mCS above 51), all methods are close and warm-up adds little.

We highlight one pattern that we return to in the discussion: on Caltech101 and UCF101, the median concept similarity (mCS) stays flat or de-

creases slightly under warm-up, even though the best concept similarity (bCS) on the same datasets rises. For completeness, we also recorded the perplexity of the generated answers, which stays almost constant across all methods (around 1.2–1.4).

4.1 Comparison with prior work on the same benchmark

Both Conti et al. [4] and Garosi et al. [7] evaluate the same models on the same datasets, which lets us situate test-time warm-up against their reported numbers. Table 2 first checks our zero-shot baseline against the Text Inclusion (TI) values reported by Conti et al.; the close match confirms that our pipeline reproduces the benchmark faithfully, so the warm-up gains are measured from a sound starting point.

Table 2: Baseline reproduction. Our zero-shot Text Inclusion (TI) on Qwen2-VL-7B-Instruct against the values reported by Conti et al. [4].

Method	C101	DTD	FLWR	PETS	U101
Conti et al. (ZS)	63.2	15.7	42.3	12.1	12.5
Ours (ZS)	63.2	15.8	40.2	12.3	12.3

Table 3 then places warm-up next to CIRCLE, the strongest open-world method of Garosi et al. CIRCLE reaches clearly higher scores, but the comparison is not like-for-like: CIRCLE iteratively builds and refines a context for the query image from model-generated predictions, whereas test-time warm-up performs label-free parameter updates using only captions of the single test image. The gap therefore reflects the advantage of explicit context refinement, while warm-up still improves over the zero-shot baseline on most datasets, most clearly on Oxford Pets and Flowers102. We omit Text Inclusion from this comparison because our Text Inclusion metric is substring-based, whereas Garosi et al. report Llama Inclusion using an LLM judge, so the two scores are not directly comparable.

4.2 GRPO post-training (preliminary)

We also ran a preliminary experiment combining the GRPO post-training phase with LoRA warm-up on Oxford Pets, using the newer Qwen2.5-VL-7B-Instruct model. Table 4 compares LoRA warm-up with and without the GRPO phase on the four benchmark metrics.

Adding the GRPO phase lowers every metric in Table 4, sharply so for Text Inclusion and best Concept Similarity, with TI falling from 64.2 to 9.0. These numbers suggest the phase simply failed, but the training dynamics² point elsewhere. Over the full run (about 105 optimisation steps), the mean

²The GRPO training runs are available at https://wandb.ai/jucamohedano/ttw_grpo_classification?nw=nwuserjucamohedano.

Table 3: Open-world Semantic Similarity (SS), best Concept Similarity (bCS), and median Concept Similarity (mCS) on Qwen2-VL-7B-Instruct. We compare our zero-shot baseline (ZS), our strongest warm-up variant (TTW-LoRA), and CIRCLE [7] with a 16-shot refined context per query. Higher is better.

Metric	Method	C101	DTD	FLWR	PETS	U101
SS	ZS	55.7	28.5	40.9	25.1	43.1
	TTW-LoRA	53.8	29.1	46.3	35.9	45.5
	CIRCLE-16	68.4	36.5	54.5	61.0	52.3
bCS	ZS	81.2	50.3	67.6	43.2	58.9
	TTW-LoRA	81.7	51.9	73.0	65.3	63.2
	CIRCLE-16	79.0	46.2	64.4	73.9	61.8
mCS	ZS	51.4	27.8	32.9	25.2	38.9
	TTW-LoRA	47.2	27.8	33.8	32.2	35.7
	CIRCLE-16	65.2	34.8	53.5	51.2	50.9

Table 4: Effect of GRPO warm-start before TTW LoRA on Oxford Pets (PETS), using Qwen2.5-VL-7B-Instruct.

Metric	TTW LoRA (no GRPO)	GRPO $\rightarrow$ TTW LoRA	Δ
TI	64.2	9.0	-55.2
SS	40.8	24.9	-15.9
bCS	76.9	41.6	-35.3
mCS	27.6	25.3	-2.3

reward rises steadily from roughly 0.29 to 0.49, and the held-out validation accuracy on Oxford Pets, measured with the trained prompt, improves from about 0.29 to 0.33 (acc@1) over the first 55 steps. The KL divergence from the reference policy grows and accelerates throughout (from about 0.001 to 0.055), and the gradient norm stays mostly below about 0.5 with intermittent spikes, the largest reaching about 2.3. So the policy drifts substantially under the small KL penalty ($\beta = 0.001$) and the run is not perfectly stable, but it does not diverge, and both the reward and the validation accuracy improve. During training, then, the model was learning the rewarded behaviour and improving its recognition of Oxford Pets under its own prompt format.

This makes the collapse in Table 4 surprising, and a diagnostic on all 3,669 Oxford Pets test images locates it after training rather than during it. We first suspected a parsing artifact, namely that the structured prompt would produce a `<think>` block and that our label extractor would fail to read the answer after `</think>`. Instead, 99.9% of the responses contain no `<think>` block at all, and almost every response reverts to the generic form “The object in the image is ...”: the structured format that GRPO was trained to produce has vanished from the merged checkpoint. The few correct hits ($\approx 10\%$) concentrate on vi-

sually distinctive breeds (for example Pug, Siamese, and Beagle), while subtle breeds that require fine-grained discrimination (for example Russian Blue, English Setter, and German Shorthaired) receive essentially none, which is the signature of chance-level captioning rather than learned classification. The contrast is the key evidence: with the trained prompt the model reached about 0.33 accuracy during training, whereas the merged checkpoint identifies the breed only about 10% of the time. We therefore read the Table 4 numbers not as a failure of GRPO, but as a downstream integration problem: the conversion and merging of the GRPO checkpoint appears to drop the trained behaviour, and a second possibility is that the subsequent per-image warm-up, which trains on generic captions, pulls the model back toward generic captioning. We could not disambiguate these within the time available, so we treat the GRPO phase as exploratory and unfinished and do not present Table 4 as evidence on whether GRPO helps open-world classification.

At test time, the GRPO checkpoint is prompted with the same structured system prompt used during training, followed by the question “What is the object in the image?”, and decoded greedily with up to 256 new tokens. The predicted label was meant to be read from the text after the `</think>` block. As the diagnostic above shows, the merged checkpoint ignores this structure, which is why the label-extraction step has almost nothing well-formed to read. The full system prompt is found in Appendix D.

5 Discussion

Test-time warm-up transfers to open-world classification. On most datasets and metrics, warm-up improves over the zero-shot baseline, and the gains are largest exactly where we would hope: on the datasets where the baseline struggles, such as Oxford Pets and Flowers102. This matches the trend reported in the original test-time warm-up paper of Rajaneesh et al. [14], where warm-up helped most on inputs that demand careful perception and added little on inputs the model already handled well. In our setting the same pattern appears as a ceiling effect: on an easy dataset like Caltech101 every method is already near the top, so there is little room for warm-up to help.

Comparing the warm-up variants. The two parameter-efficient variants, LoRA and SVF, give larger gains than dense finetuning. We are cautious about attributing this to the parameterization alone, because the parameter-efficient variants were also trained for more epochs (five versus two), so part of their advantage may simply reflect longer adaptation rather than the choice of trainable parameters. What is consistent across all three is the dataset-level pattern above: warm-up helps most on the perception-heavy, fine-grained datasets and least where the baseline is already strong.

Best versus median similarity, and the granularity gap. A more cautious reading comes from comparing bCS and mCS. On Caltech101 and UCF101,

warm-up raises the best concept similarity (bCS) while the median (mCS) stays flat or falls. Garosi et al. [7] observe that bCS can be inflated when a model emits a broad, ungrounded list of candidate concepts, because only the single concept closest to the ground truth needs to match, whereas mCS takes the median over all extracted concepts and therefore reflects the most typical concept the model predicts. The divergence we see is consistent with this: on inputs the zero-shot model already handles comparatively well, warm-up makes the model propose a wider set of concepts, raising the best match without making its typical concept any more precise. We therefore treat bCS gains with caution and regard mCS as the more conservative signal of genuine improvement.

We read this same pattern as the central limitation of warm-up, and it becomes clearest when we place warm-up next to CIRCLE (Table 3). Warm-up improves most where the zero-shot model is weak, especially Oxford Pets, but it does not close the gap to CIRCLE on SS or mCS. The difficulty is therefore not only making the model attend to the image, but making it answer at the right level of specificity: after warm-up the model may correctly notice that the image contains a dog, yet still fail to produce the breed name expected by Oxford Pets. This is the bCS–mCS story again, now across methods rather than within one. Warm-up widens and grounds what the model sees, which lifts the best match, but it has no signal telling it whether the expected answer should be broad (“dog”) or specific (“English Setter”), so the typical answer stays at the wrong granularity. CIRCLE addresses precisely this: in the open-world setting Garosi et al. [7] use in-context images and iterative pseudo-label refinement to let the model infer the semantic granularity of the task, whereas TTW adapts only from captions of the single test image and has no such external reference. This explains why CIRCLE is much stronger on mCS, for example on Oxford Pets (51.2 versus 32.2 for TTW-LoRA), while on bCS warm-up can look competitive on some datasets only because of the inflation effect above, not because its predictions are more precise. Overall, label-free per-image adaptation recovers part of the benefit of test-time context, but not the precise, grounded predictions that in-context refinement provides. We do not compare against the original test-time warm-up paper because it evaluates visual question answering with plain accuracy, whereas open-world classification requires the inclusion and similarity metrics that Conti et al. [4] designed for free-form predictions.

Reading the two model generations. Finally, a caveat on how the result tables relate. Our cross-method comparison (Table 1) uses Qwen2-VL-7B-Instruct, whereas the GRPO experiment uses the newer Qwen2.5-VL-7B-Instruct. The two model generations differ in strength on this benchmark, so numbers from the Qwen2-VL-7B-Instruct table should not be compared cell by cell with the Qwen2.5-VL-7B-Instruct GRPO results, i.e., each should be read against its own baseline.

6 Limitations

A first limitation comes from the task itself. Without a fixed label set, the same image can have several valid answers at different levels of granularity: for example, it might be reasonable to describe the same image as a “pet”, a “dog”, or a “Labrador”. This makes evaluation inherently ambiguous, because there is no single exact label against which every prediction can be judged. The benchmark addresses this problem with soft matching metrics rather than plain accuracy, but this also means that improvements should be interpreted relative to those metrics, not as absolute measures of correctness.

A second limitation is computational cost. Test-time warm-up is expensive. For every test image, it keeps the model in memory and runs several optimization steps, including gradients and optimizer state during adaptation. Pairing TTW with a GRPO warm-start increases this cost further, because it adds a separate reinforcement-learning stage before test-time adaptation begins. If more compute is available, some of the same gains might be achieved more simply by using a larger or stronger base model without per-image updates. In this sense, TTW trades inference-time compute for model capacity, and it is most attractive when using a larger model is not feasible.

GRPO is another limitation of our study. We included it as an exploratory warm-start before test-time warm-up, and its training dynamics were encouraging: as shown in Section 4.2, both the reward and the held-out Oxford Pets validation accuracy improved during training. However, the checkpoint we evaluated had lost the structured format that GRPO was trained to produce, so the negative numbers in Table 4 reflect a downstream integration failure rather than the effect of GRPO itself. The experiment is unfinished: we have evidence that training was working and that the evaluated model had collapsed, but not yet a confirmed fix, so we draw no conclusions about GRPO’s effect on open-world classification from these numbers.

A related point concerns what we optimised. We used GRPO mainly to encourage a structured reasoning format before classification, not to directly optimise the open-world metrics used by the benchmark. That validation accuracy rose alongside the reward is mild evidence that, at least early in training, the rewarded behaviour tracked real improvement rather than being gamed. We remain cautious, though: the policy drifted a long way from the reference model, with the KL term growing by more than an order of magnitude under a small penalty, and the late part of training, where drift is largest, is exactly the regime in which reward hacking can appear. We did not explicitly test for it, and the mismatch between what we rewarded (format) and what the benchmark measures (open-world recognition) is reason enough to revisit the reward design and to check for hacking before relying on the GRPO phase.

A more promising direction would be to design the reward around the actual open-world classification objective, especially the trade-off between specificity and correctness, and to verify that the trained behaviour survives checkpoint conversion and the subsequent warm-up before evaluating it. Another direction is to study whether the required fine-grained knowledge is already present in

the model, and how this changes with model scale. Larger models may encode more of the specialized concepts that datasets such as Oxford Pets require, which could make both GRPO and test-time warm-up less useful since it might become achievable with sufficient test-time compute.

A Dataset statistics

Table 5: Datasets used in this work, grouped by category following Conti et al. [4]. We evaluate on five of the ten datasets in their benchmark. Sizes and class counts are taken from their Table 3; dataset references are cited in the dataset column.

Category	Dataset	Images	Classes
Prototypical	Caltech101	2,465	100
Non-prototypical	DTD	1,692	47
	UCF101	3,783	101
Fine-grained	Flowers102	2,463	102
	Oxford Pets	3,669	37

B Auxiliary warm-up prompts

The ten auxiliary prompts used to generate warm-up captions are listed below, verbatim.

Auxiliary warm-up prompts

1. What is happening in this image?
2. Describe the main subject of this image in detail.
3. What objects or people are visible in this image?
4. What actions are the subjects performing in this image?
5. What does the background reveal about this image?
6. What is unusual or unique about this image?
7. What details in this image might someone easily overlook?
8. Are there any signs, symbols, or text in this image? If so, what do they say?
9. Explain the possible relationships or roles of the people, animals, or objects in this scene. What hints or clues suggest these relationships?
10. Based on visual cues, infer what might have happened just before and what might happen right after this image was captured.

C Test-time warm-up algorithm

Algorithm 1 Test-Time Warm-up (TTW)

Require: Test image x , model f_θ , auxiliary prompts $\mathcal{Q} = \{q_i\}_{i=1}^N$

Ensure: Predicted class $\hat{y}$

```
1:  $\mathcal{D} \leftarrow \emptyset$ 
2: for all  $q_i \in \mathcal{Q}$  do
3:   Generate candidate captions  $f_\theta(x, q_i)$   $\triangleright$  model captions its own input
   image
4:   Score each candidate against  $x$  with CLIP
5:    $c_i \leftarrow$  highest-scoring caption
6:    $\mathcal{D} \leftarrow \mathcal{D} \cup \{(x, q_i, c_i)\}$   $\triangleright$  warm-up example: image + prompt  $\rightarrow$  caption
7: end for
8:  $\theta_{\text{orig}} \leftarrow \theta$   $\triangleright$  snapshot trainable parameters
9:  $\theta' \leftarrow$  Fine-tune  $f_\theta$  on  $\mathcal{D}$  for  $E$  epochs  $\triangleright$  updated parameters  $\theta'$ 
10:  $\hat{y} \leftarrow f_{\theta'}(x)$   $\triangleright$  classify with the warmed-up model
11: return  $\hat{y}$ 
```

D GRPO prompt template

At inference, the GRPO checkpoint receives the following system prompt, followed by the user question “What is the object in the image?”.

GRPO training and inference prompt

You are an expert visual reasoner. For every image,
first think through what you see inside <think>...</think>,
→ using three sub-tags:
<HasProperty> for visible properties, <HasA> for visible
→ parts, and
<AtLocation> for visible setting or context.

Use this exact structure:

```
<think>
<HasProperty>2-5 short, comma-separated visible
→ properties</HasProperty>
<HasA>2-5 short, comma-separated visible parts</HasA>
<AtLocation>2-5 short, comma-separated visible settings or
→ places</AtLocation>
Reason on the attributes extracted to answer the user's
→ question after thinking.
```

```
</think>
answer the user's question directly
```

E Hyperparameters

Table 6: Test-time warm-up hyperparameters for the three finetuning methods.

Parameter	Dense	LoRA	SVF
Learning rate	10^{-6}	10^{-4}	10^{-4}
Epochs	2	5	5
Batch size	5	5	5

Table 7: Settings shared across all warm-up methods.

Parameter	Value
Precision	<code>bfloat16</code>
Evaluation batch size	1
CLIP model (caption scoring)	<code>clip-vit-large-patch14-336</code>
Candidate captions per prompt	10
Caption sampling temperature	0.75
Caption max new tokens	128
Random seed	30

Table 8: GRPO post-training hyperparameters (ver1 framework).

Category	Parameter	Value
Model & adaptation	Base model	Qwen2.5-VL-7B-Instruct
	Adaptation	LoRA (rank 64, $\alpha=32$)
Data	Train batch size	128
	Max prompt / response	2048 / 4096 tokens
	Rollouts per prompt	5
Optimization	Actor learning rate	1e−6
	PPO mini / micro batch	64 / 8
	Total epochs	5
	KL penalty	$\beta=0.001$ with <code>low_var_kl</code>
	Entropy coefficient	0
	Advantage estimator	GRPO
Infrastructure	GPUs	4, single node
	Rollout engine	vLLM, tensor parallel size 4
	GPU memory utilization	0.4
Reward	Format score	0.30
	Answer score	0.70

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